



B.Tech. Degree IV Semester Regular/Supplementary Examination June 2022

SE 19-206-0402 HEAT AND MASS TRANSFER OPERATIONS (2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

- CO1: Explain different modes of heat transfer and carry out the process calculations in various geometries.
 CO2: Recognise the basic methodology in designing heat exchangers and evaporators.
 CO3: Estimate diffusion coefficients and convective mass transfer rates.
 CO4: Select and design the most suitable equipment for absorption, distillation, liquid-liquid extraction, and solid-liquid extraction.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze,
 L5 – Evaluate, L6 – Create
 PO – Programme Outcome

**PART A**(Answer *ALL* questions)

		(8 × 3 = 24)	Marks	BL		
I.				L2	1	1,2
(a)	State Fourier's law of heat conduction and illustrate the steady state and unsteady state heat transfer through a furnace wall.		3	L2	1	1,2
(b)	Explain various regimes of boiling with the help of a boiling curve.		3	L1	2	1
(c)	Describe different types of feeding arrangements in multiple effect evaporators.		3	L1	2	1
(d)	Draw the temperature –length curves for counter current and parallel flow heat exchange and derive the equation for LMTD.		3	L2	3	1,2
(e)	What are various forms of mass transfer theories and explain each of them?		3	L1	3	1
(f)	What is absorption? Give a few criteria for selecting a suitable absorbent.		3	L1	4	1
(g)	Define relative volatility and explain its significance in distillation.		3	L1	4	1,2
(h)	Define bound moisture, unbound moisture and equilibrium moisture in solids.		3			

PART B

(4 × 12 = 48)

II.	(a)	Derive an expression for the steady state heat conduction through a cylindrical wall.	6	L2	1	1
	(b)	A hollow cylinder of length of 1 m, inner radius 0.01 m and outer radius 0.05 m has a thermal conductivity (k) 20 W/mK. The inside surface temperature is at 500 K. What is the outside surface temperature of the cylinder when the rate of radial heat transfer is 3.13 kW?	6	L5	1	1,2,3

OR

III.	(a)	Differentiate between natural convection and forced convection.	4	L4	1	1
	(b)	Define individual and overall heat transfer coefficient. Also derive an expression for the overall heat transfer coefficient based on outside area.	8	L1	1	1,2

(P.T.O)

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IV.	A counter flow heat exchanger is used to cool 2200 kg/hr of oil ($c_p=2.5 \text{ kJ/kgK}$), from 100°C to 35°C by the use of water entering at 17°C . If the overall heat transfer coefficient is expected to be $1.5 \text{ kW/m}^2\text{K}$, make calculations for the water flow rate, the surface area required and the effectiveness of heat exchanger. Presume that the exit temperature of water is not to exceed 85°C . Use NTU-effectiveness approach.	12	L3	2	1,3
OR					
V. (a)	What are different types of evaporators? Explain any one type in detail.	6	L1	3	1,2
(b)	A single-effect evaporator is used to concentrate 7 kg/s of a solution from 10 to 50 per cent of solids. Steam is available at 205 kN/m^2 and evaporation takes place at 13.5 kN/m^2 . If the overall heat transfer coefficient is $3 \text{ kW/m}^2\text{K}$, calculate the heating surface required and the amount of steam used if the feed to the evaporator is at 294 K and the condensate leaves the heating space at 352.7 K . The specific heat capacity of a 10 per cent solution is 3.76 kJ/kgK , the specific heat capacity of a 50 per cent solution is 3.14 kJ/kg K . The following data are given: Enthalpy of steam at $394\text{K}=2530\text{kJ/kg}$ Enthalpy of steam at $325\text{K}=2594\text{kJ/kg}$	6	L3	3	1,2
VI. (a)	Derive an equation for molar flux of diffusion through a stagnant gas film.	6	L2	4	1
(b)	Calculate the rate of diffusion of butanol at 20°C under unidirectional steady state conditions through a 0.1 cm thick film of water when the concentrations of butanol at the opposite sides of the film are 10% and 4% respectively (by weight). The diffusivity of butanol in water solution is $5.9 \times 10^{-6} \text{ cm}^2/\text{sec}$. The densities of 10% and 4% solutions are 0.971 and 0.992 g/cm^3 respectively. Molecular weight of butanol is 74 and that of water is 18.	6	L5	4	1,2
OR					
VII. (a)	Summarise the causes of loading and flooding in packed towers.	4	L2	4	1,2
(b)	It is desired to absorb 90% of acetone from a gas containing 1 mol% acetone in air in a counter current stage tower. The total inlet gas to the tower is 30 kmol/h , the total inlet pure water flow to be used to absorb acetone is 90 kmol/h . The process is to operate isothermally at 300 K and a total pressure of 1 atm . The equilibrium relation is $y = 2.53x$, where x and y denote the mole fractions of acetone in liquid and vapour phases. Determine the number of theoretical stages required for this operation.	8	L3	4	1,2
VIII. (a)	Explain azeotropic and extractive distillation.	5	L2	4	1
(b)	A mixture of 40 mol% benzene and 60 mol% toluene is being flash distilled at a rate of 10 kmol/h at 1 atm total pressure. The liquid product should not contain more than 30 mol % benzene. Calculate the amounts and the compositions of the top and the bottom products. The relative volatility of benzene in the mixture is 2.5.	7	L5	4	1,2
OR					
IX. (a)	It is desired to extract oil from soyabean seeds. Suggest a suitable extractor and explain its working.	6	L3	4	1,3
(b)	Explain the working of a spray dryer.	6	L2	4	1,3

Blooms's Taxonomy Levels

L1 -- 24.1%, L2 -- 26.6%, L3 -- 29.3%, L4 -- 20%